



CIS 635 Knowledge Discovery & Data Mining

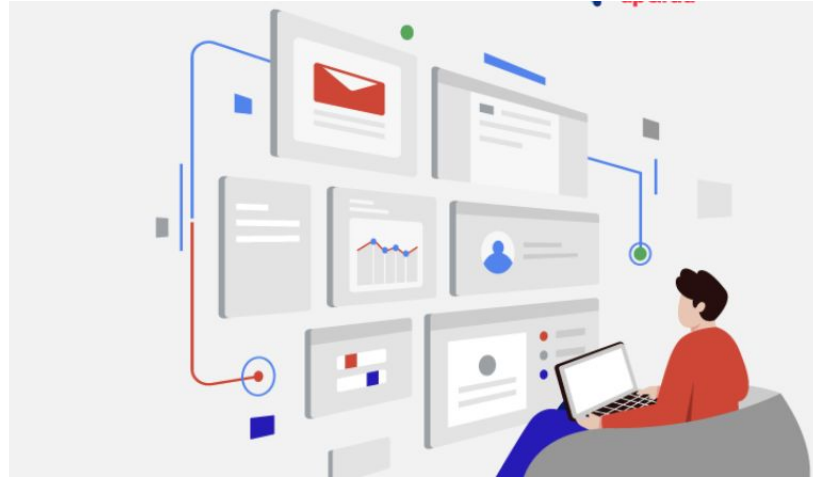
Data modalities & Data Types - Introduction and Overview



Outline

- Everyday observed data and their modalities

Everyday observed (real life) data



Everyday observed (real life) data



- We have seen an **explosion of data** in recent days, especially in the **last few years!**

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- The transition from the Analog to the Digital World has made this Big Shift (was unimaginable a few decades ago)

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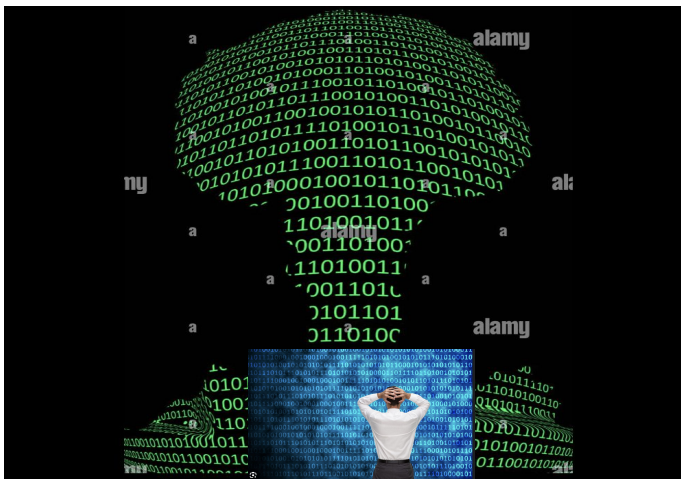


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- It's very difficult to **segregate them into different types** as data come in different **format** and in different **modalities**.
- We will try a **simplified approach** to **group** them and try to cover most (**NOT ALL**) data we perceive in our everyday life.



Everyday observed (real life) data

- Structured data
- Un(and Semi) structured data

Structured data

Loan Approval Data Set

Data Card Code (2) Discussion (0)

Loan_Train.csv (38.01 kB)

Detail Compact Column 10 of 13 columns

Loan_ID	Gender	Married	Dependents	Education	Self_Employed	ApplicantIncome
614 unique values	Male 80%	398 65% true	0 56%	Graduate 78%	82 13% true	150 8
	Female 18%	213 35% false	1 17%	Not Graduate 22%	500 81% false	
	Other (13) 2%	3 0% [null]	Other (167) 27%		32 5% [null]	
LP001002	Male	No	0	Graduate	No	5849
LP001003	Male	Yes	1	Graduate	No	4583
LP001005	Male	Yes	0	Graduate	Yes	3000
LP001006	Male	Yes	0	Not Graduate	No	2583
LP001008	Male	No	0	Graduate	No	6000
LP001011	Male	Yes	2	Graduate	Yes	5417
LP001013	Male	Yes	0	Not Graduate	No	2333
LP001014	Male	Yes	3+	Graduate	No	3036
LP001018	Male	Yes	2	Graduate	No	4006

- Generally organized in **tables** and collected through filling **forms** (manual or online)
- Stored in **databases/spread-sheets** mainly
- Also popular the **.csv** file format

- Opening a bank account
- University registration
- Gmail
- Amazon account
- Your health profile

[Kaggle loan approval dataset](#)

Structured data

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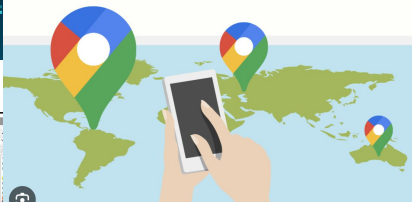
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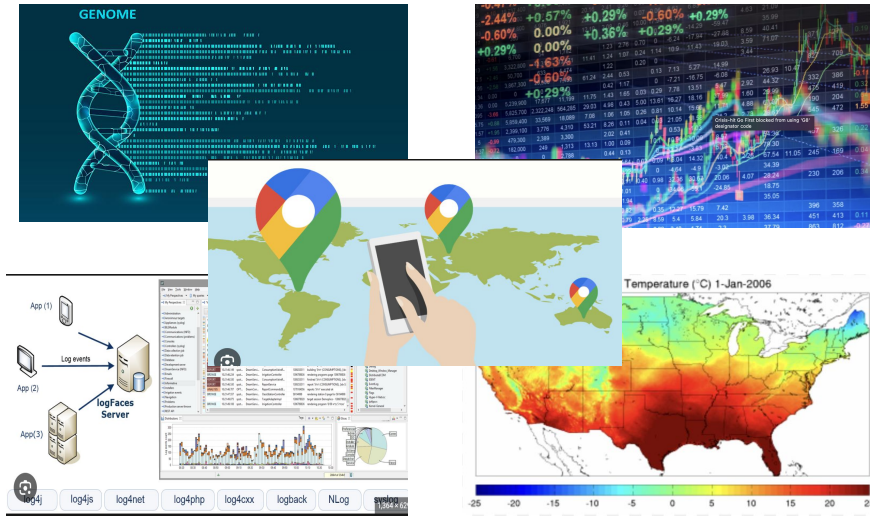
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- Opening a **bank account**
- **University registration**
- **Gmail, Azure, and/or Amazon** account
- Your immigration, health, social media **profile**

[Kaggle loan approval dataset](#)



Structured data

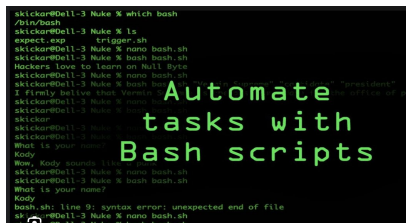


- There also other formats, or you can convert them to
- Software generated logs
- Some are stand alone, white others are sequences and/or series

- Genomics data
- Stock prices
- Your CC history
- Weather data
- Google maps



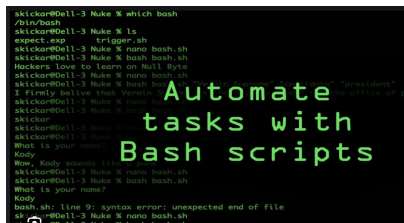
- Free forms
- Stored in data lake/warehouses mainly
- **Languages:** sequence of strings (semantics)
- **Audio:** Language + Acoustics; Music
- **Image:** Visual representation of the world
- **Video:** Sequence of images



Un/Semi Structured data



- Free forms
- Stored in data lake mainly?
- Languages: sequence of strings
- Audio: Language + Acoustics; Music
- Image : Visual representation of the world
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- Some are sequences, while others are stand alone
- Social media data (emotions, vives)

Un/Semi Structured data



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- Some are sequence, while others are stand alone
- **Social media** data (discussions, messages, emotions, vibes)



QA